



**NATIONAL
SENIOR CERTIFICATE/
NASIONALE
SENIOR SERTIFIKAAT**

GRADE/GRAAD 11

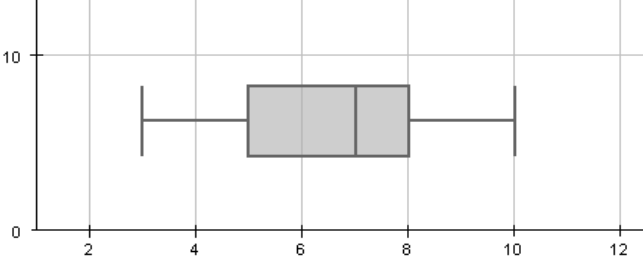
NOVEMBER 2021

**MATHEMATICS P2/WISKUNDE V2
MARKING GUIDELINE/NASIENRIGLYN**

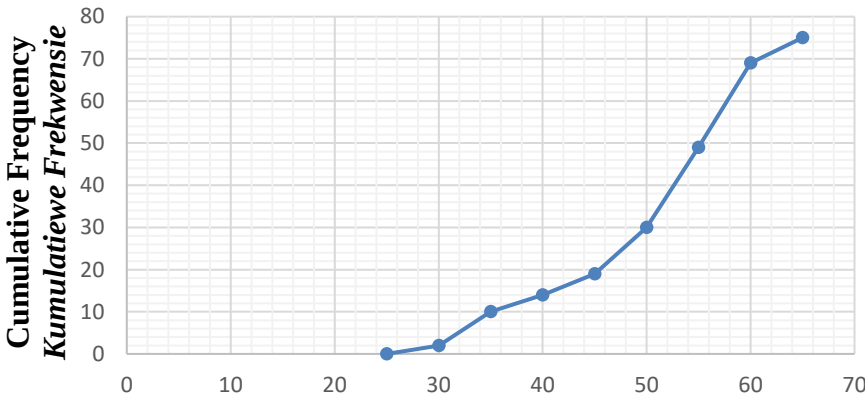
MARKS/PUNTE: 150

This marking guideline consists of 16 pages./
Hierdie nasienriglyn bestaan uit 16 bladsye.

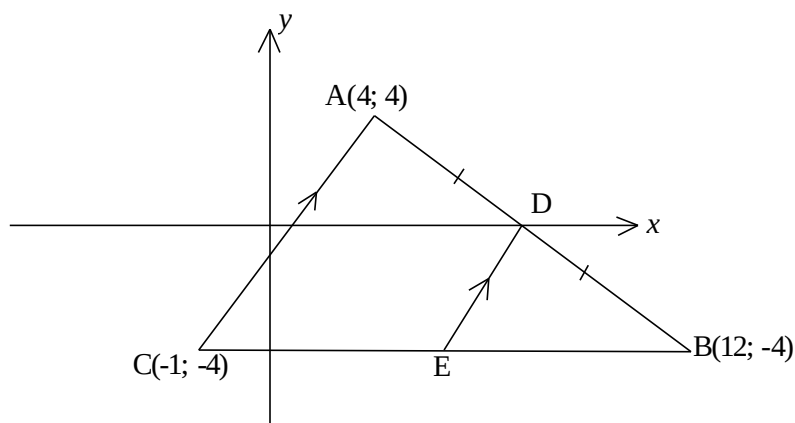
QUESTION 1/VRAAG 1

1.1	$\bar{x} = 6,4$	✓ answer / antwoord	(1)
1.2	Min/Min = 3; Max./Maks. = 10; Lower Quartile/Onderste kwartiel = 5; Median/Mediaan = 7; Upper Quartile/Boonste kwartiel = 8	✓ for both min. and max. vir beide min. en maks. ✓ for the rest / vir die res	(2)
1.3		✓ for the box / vir die mond ✓ whiskers / snorre	(2)
1.4	Skewed to the left, mean is smaller than the median OR longer rectangle on the left. <i>Skeef na links, Gemiddelde is kleiner as mediaan OF langer reghoek op linkerkant.</i>	✓ answer / antwoord ✓ reason / rede	(2)
1.5	Standard deviation/Standaardafwyking = 2, 01	✓✓ answer / antwoord	(2)
1.6	Slept well, if bigger than 8,61. / <i>Het goed geslaap, indien groter as 8,61.</i> Answer 1 learner./ Antwoord 1 leerder	✓ 8,61 ✓ answer / antwoord	(2)
			[11]

QUESTION 2/VRAAG 2

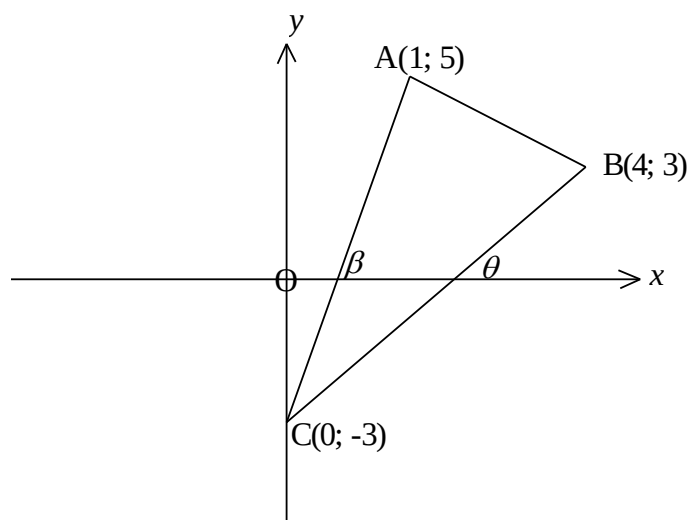
2.1	<table><tr><th>Age <i>Ouderdom</i></th><th>Frequency <i>Frekwensie</i></th><th>Cumulative Frequency <i>Kumulatiewe Frekwensie</i></th></tr><tr><td>$25 < A \leq 30$</td><td>2</td><td>2</td></tr><tr><td>$30 < A \leq 35$</td><td>8</td><td>10</td></tr><tr><td>$35 < A \leq 40$</td><td>4</td><td>14</td></tr><tr><td>$40 < A \leq 45$</td><td>5</td><td>19</td></tr><tr><td>$45 < A \leq 50$</td><td>11</td><td>30</td></tr><tr><td>$50 < A \leq 55$</td><td>19</td><td>49</td></tr><tr><td>$55 < A \leq 60$</td><td>20</td><td>69</td></tr><tr><td>$60 < A \leq 65$</td><td>6</td><td>75</td></tr></table>	Age <i>Ouderdom</i>	Frequency <i>Frekwensie</i>	Cumulative Frequency <i>Kumulatiewe Frekwensie</i>	$25 < A \leq 30$	2	2	$30 < A \leq 35$	8	10	$35 < A \leq 40$	4	14	$40 < A \leq 45$	5	19	$45 < A \leq 50$	11	30	$50 < A \leq 55$	19	49	$55 < A \leq 60$	20	69	$60 < A \leq 65$	6	75	✓ for first 4 vir eerste 4 ✓ last 4 laaste 4	(2)
Age <i>Ouderdom</i>	Frequency <i>Frekwensie</i>	Cumulative Frequency <i>Kumulatiewe Frekwensie</i>																												
$25 < A \leq 30$	2	2																												
$30 < A \leq 35$	8	10																												
$35 < A \leq 40$	4	14																												
$40 < A \leq 45$	5	19																												
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$50 < A \leq 55$	19	49																												
$55 < A \leq 60$	20	69																												
$60 < A \leq 65$	6	75																												
2.2	Ages of Teachers <i>Ouderdomme van Onderwysers</i> 	✓ start point beginpunt ✓ end point eindpunt ✓✓ shape vorm	(4)																											
2.3	Median / <i>Mediaan</i> = 52 (Accept/ <i>Aanvaar</i> 51 – 53)	✓✓ answer / <i>antwoord</i>	(2)																											
2.4	Percentage / <i>Persentasie</i> = $\frac{75 - 57}{75} = \frac{18}{75} = 24\%$	✓ calculation / <i>berekening</i> ✓ answer / <i>antwoord</i>	(2)																											
			[10]																											

QUESTION 3/VRAAG 3



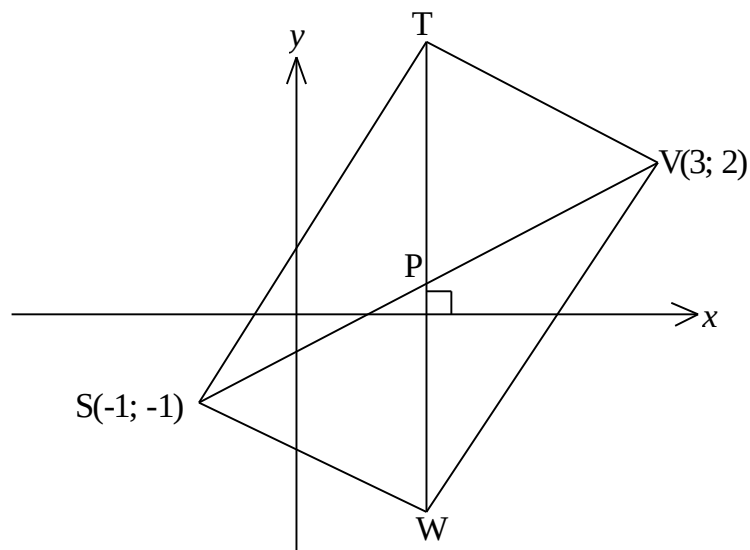
3.1	$AB = \sqrt{(12 - 4)^2 + (-4 - 4)^2}$ $AB = 11,31 \text{ units / eenhede}$		✓ substitution / <i>vervanging</i> ✓ answer / <i>antwoord</i>	(2)
3.2	$D\left(\frac{4 + 12}{2}; \frac{4 - 4}{2}\right)$ $= D(8; 0)$		✓ substitution / <i>vervanging</i> ✓ answer / <i>antwoord</i>	(2)
3.3	$m = \frac{4 - (-4)}{4 - (-1)} = \frac{8}{5}$ $y = mx + c$ $0 = \frac{8}{5} \times 8 + c$ $c = -\frac{64}{5}$ $\therefore y = \frac{8}{5}x - \frac{64}{5}$	$m = \frac{4 - (-4)}{4 - (-1)} = \frac{8}{5}$ $y - y_1 = m(x - x_1)$ $y - 0 = \frac{8}{5}(x - 8)$ $y = \frac{8}{5}x - \frac{64}{5}$	✓ gradient / <i>gradiënt</i> ✓ substitution / <i>vervanging</i> ✓ y-intercept / <i>y-afsnit</i> ✓ answer / <i>antwoord</i>	(4)
3.4	$y = \frac{8}{5}x - \frac{64}{5}$ $-4 = \frac{8}{5}x - \frac{64}{5}$ $\therefore x = \frac{11}{5}$ $E\left(\frac{11}{2}; -4\right)$ OR / OF $E\left(\frac{11}{2}; -4\right)$ (Midpoint Theorem) / (<i>Middelpunt-Stelling</i>)		✓ substitution -4 <i>vervanging -4</i> ✓ value of x <i>waarde van x</i> ✓ coordinates of E <i>koördinate van E</i> OR / OF ✓ x – value / <i>waarde</i> ✓ y – value / <i>waarde</i> ✓ reason / <i>rede</i>	(3)
				[11]

QUESTION 4/VRAAG 4



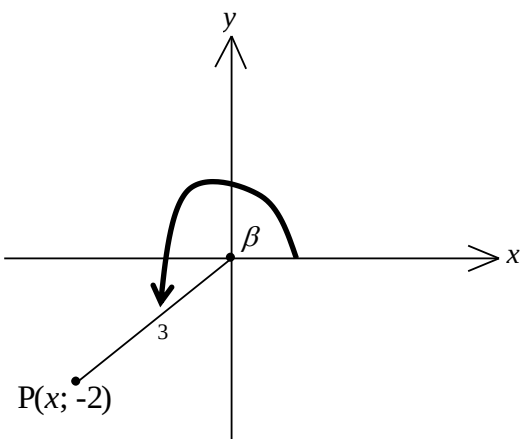
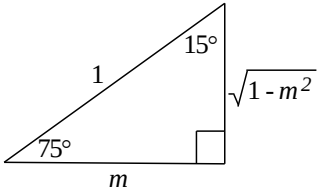
4.1	D(-3; -1)	✓ x-coordinate / <i>koördinaat</i> ✓ y-coordinate / <i>koördinaat</i>	(2)
4.2	$FC = \sqrt{(8-0)^2 + (p+3)^2} = 12$ $64 + p^2 + 6p + 9 = 144$ $p^2 + 6p - 71 = 0$ Using the quadratic formula: $p = 5,94$ or $p = -11,94$ $p = 6$ or $p = -12$	✓ substitution into the formula and equating to 12 <i>vervanging in die formule en stel gelyk aan 12</i> ✓ squaring both sides <i>kwadreer albei kante</i> ✓ standard form/standaardvorm ✓ $p = 6$ ✓ $p = -12$	(5)
4.3	$m_{BC} = \frac{3 - (-3)}{4 - 0} = \frac{3}{2}$ $\therefore \tan \theta = \frac{3}{2}$ $\therefore \theta = 56,31^\circ$ $\tan \beta = 8$ $\therefore \beta = 82,87^\circ$ $\therefore \widehat{ACB} = 82,87^\circ - 56,3^\circ$ $\therefore \widehat{ACB} = 26,56^\circ$	✓ gradient of BC $\checkmark \tan \theta = \frac{3}{2}$ ✓ for θ ✓ for β ✓ for $\widehat{ACB}$	(5)
			[12]

QUESTION 5/VRAAG 5



	$P\left(\frac{-1+3}{2}; \frac{-1+2}{2}\right)$ $= P\left(1; \frac{1}{2}\right)$ $SV = \sqrt{(3+1)^2 + (2+1)^2}$ $SV = 5 \text{ units / eenhede}$ $TW = 5 \text{ units / eenhede (diagonals of a rectangle)}$ $\text{(hoeklyne van 'n reghoek)}$ $T(1; 3)$ $W(1; -2)$	✓ coordinates of P <i>koördinate van P</i> ✓ substitution / <i>vervanging</i> ✓ for/vir SV ✓ for/vir TW ✓ coordinates of T <i>koördinate van T</i> ✓ coordinates of W <i>koördinate van W</i>	(6)
			[6]

QUESTION 6/VRAAG 6

6.1	$\sin \beta = -\frac{2}{3}$  $x^2 = (3)^2 - (-2)^2$ $x = -\sqrt{5}$ $\therefore 1 + \tan^2 \beta$ $= 1 + \left(\frac{-2}{-\sqrt{5}} \right)^2$ $= \frac{9}{5}$	✓ for solving for sin oplos vir sin ✓ sketch in the correct quad skets in die korrekte kwadrant ✓ value of x/ waarde van x ✓ substitution / vervanging ✓ answer / antwoord	(5)
6.2			
6.2.1	$\cos^2 105^\circ$ $= [\cos(180^\circ - 75^\circ)]^2$ $= (-\cos 75^\circ)^2$ $= m^2$	✓ for/vir - cos 75° ✓ for/vir m ²	(2)
6.2.2	$\sin 15^\circ$ $= \cos 75^\circ$ $= m$	✓ for/vir cos 75° ✓ for/vir m	(2)
6.2.3	$\tan 15^\circ = \frac{m}{\sqrt{1-m^2}}$ OR / OF $\tan 15^\circ = \frac{\sin 15^\circ}{\cos 15^\circ} = \frac{\cos 75^\circ}{\sin 75^\circ} = \frac{m}{\sqrt{1-m^2}}$	✓✓ for correct answer only vir korrekte antwoord (accuracy / akkuraatheid)	(2)

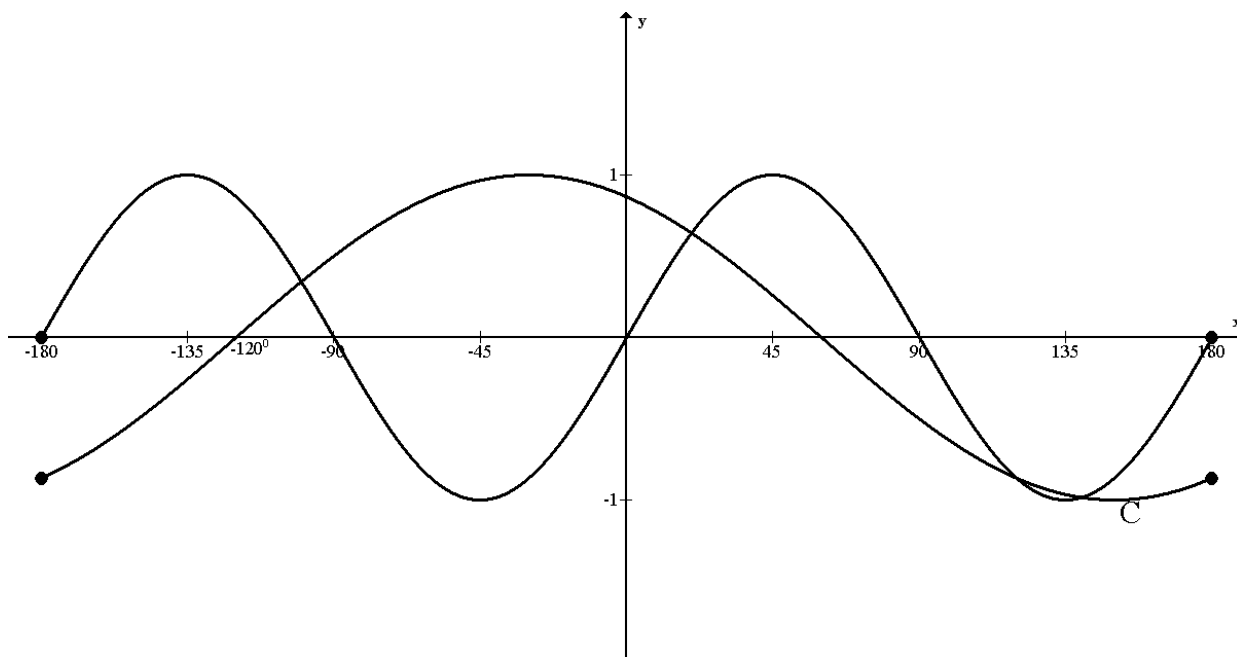
6.3.1	$\frac{\cos(180^\circ - k) \cdot \sin(k - 90^\circ) - 1}{\tan^2(540^\circ + k) \cdot \sin(90^\circ + k) \cdot \cos(-k)}$ $= \frac{-\cos k \cdot -\cos k - 1}{\tan^2 k \cdot \cos k \cdot \cos k}$ $= \frac{\cos^2 k - 1}{\frac{\sin^2 k}{\cos^2 k} \cdot \cos^2 k}$ $= \frac{-\sin^2 k}{\sin^2 k}$ $= -1$	✓ for/vir $-\cos k$ ✓ for/vir $-\cos k$ ✓ for/vir $\tan^2 k$ ✓ for/vir $\cos k$ ✓ for/vir $\cos k$ ✓ for changing $\tan^2 k$ vir verandering van $\tan^2 k$ ✓ for answer / vir antwoord	(7)
6.3.2	$\tan^2 k \cdot \cos^2 k = 0$ $\tan^2 k = 0 \text{ or/of } \cos^2 k = 0$ $\tan k = 0 \text{ or/of } \cos k = 0$ $\therefore k = 0^\circ \text{ or/of } k = 90^\circ \text{ or/of } k = 180^\circ$ $\text{or/of } k = 270^\circ \text{ or/of } k = 360^\circ$	✓ $\tan^2 k \cdot \cos^2 k = 0$ $\tan^2 k = 0$ or/of $\cos^2 k = 0$ ✓✓ $\tan k = 0$ or/of $\cos k = 0$ ✓ $k = 0^\circ$; $k = 90^\circ$ ✓ $k = 180^\circ$; $k = 270^\circ$ ✓ $k = 360^\circ$	(6)
6.4	$\frac{1 + \sin \theta}{1 - \sin \theta} - \frac{1 - \sin \theta}{1 + \sin \theta} = \frac{4 \tan \theta}{\cos \theta}$ $\text{LHS/LK} = \frac{1 + \sin \theta}{1 - \sin \theta} - \frac{1 - \sin \theta}{1 + \sin \theta}$ $= \frac{1 + 2 \sin \theta + \sin^2 \theta - 1 + 2 \sin \theta - \sin^2 \theta}{(1 - \sin \theta)(1 + \sin \theta)}$ $= \frac{4 \sin \theta}{1 - \sin^2 \theta}$ $= \frac{4 \sin \theta}{\cos^2 \theta}$ $= \frac{4 \tan \theta}{\cos \theta} = \text{RHS/RK}$	✓ for/vir LCD/KGD ✓ for/vir $1 + 2 \sin \theta + \sin^2 \theta$ ✓ for/vir $-1 + 2 \sin \theta - \sin^2 \theta$ ✓ for/vir $\frac{4 \sin \theta}{1 - \sin^2 \theta}$ ✓ for/vir $\cos^2 \theta$	(5)

6.5	$6\sin^2 \theta + \cos \theta = 4$ $6(1 - \cos^2 \theta) + \cos \theta = 4$ $6 - 6\cos^2 \theta + \cos \theta - 4 = 0$ $-6\cos^2 \theta + \cos \theta + 2 = 0$ $6\cos^2 \theta - \cos \theta - 2 = 0$ $(3\cos \theta - 2)(2\cos \theta + 1) = 0$ $\cos \theta = \frac{2}{3} \quad \text{or/of} \quad \cos \theta = -\frac{1}{2}$ $\theta = 48,19^\circ + 360^\circ.k \quad \text{or/of} \quad \theta = 311,81^\circ + 360^\circ.k$ OR/OF $\theta = 120^\circ + 360^\circ.k \quad \text{or/of} \quad \theta = 240^\circ + 360^\circ.k$ where/waar $k \in \mathbb{Z}$	✓ for/vir $1 - \cos^2 \theta$ ✓ for the two general solutions/ <i>vir die twee algemene oplossings</i> ✓ for standard form/ <i>vir standaardvorm</i> ✓ for factors / <i>vir faktore</i> ✓ for the other two general solutions/ <i>vir die ander twee algemene oplossings</i> ✓ for answers of $\cos \theta = \frac{2}{3}$ <i>vir antwoorde van $\cos \theta = \frac{2}{3}$</i> ✓ for answers of $\cos \theta = -\frac{1}{2}$ <i>vir antwoorde van $\cos \theta = -\frac{1}{2}$</i>	(7)
6.6	$p = \tan A + \sin A \quad \text{and/en} \quad q = \tan A - \sin A$ $pq = (\tan A + \sin A)(\tan A - \sin A)$ $pq = \tan^2 A - \sin^2 A$ $pq = \frac{\sin^2 A}{\cos^2 A} - \sin^2 A$ $pq = \frac{\sin^2 A - \sin^2 A \cos^2 A}{\cos^2 A}$ $pq = \frac{\sin^2 A(1 - \cos^2 A)}{\cos^2 A}$ $pq = \tan^2 A \sin^2 A$	✓ for substitution <i>vir vervanging</i> ✓ for multiplication <i>vir vermenigvuldiging</i> ✓ for changing $\tan^2 A$ <i>vir verandering van $\tan^2 A$</i> ✓ for the numerator <i>vir die teller</i> ✓ for the common factor <i>vir die gemene faktor</i>	(5)
			[41]

QUESTION 7/VRAAG 7

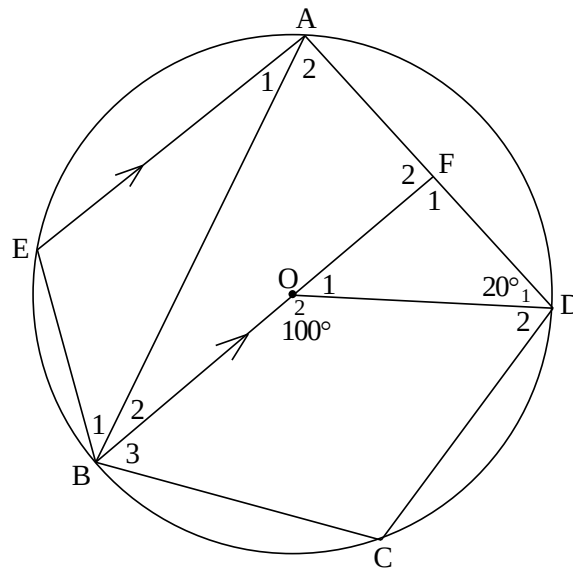
7.1	Exterior angle of a triangle / <i>Buitehoek van 'n driehoek</i>	✓ answer / <i>antwoord</i>	(1)
7.2	$\frac{PS}{\sin 29^\circ} = \frac{10}{\sin 11^\circ}$ $PS = \frac{10 \cdot \sin 29^\circ}{\sin 11^\circ}$ $PS = 25,41 \text{ m}$	✓ for sub. in sine rule <i>vir verv. in sinus-reël</i> ✓✓ for the answer <i>vir die antwoord</i>	(3)
7.3	$PQ^2 = (30)^2 + (25,41)^2 - 2 \times 30 \times 25,41 \times \cos 40^\circ$ $PQ = 19,44 \text{ m}$ $\frac{\sin Q_1}{25,41} = \frac{\sin 40^\circ}{19,44}$ $\sin Q_1 = \frac{25,41 \times \sin 40^\circ}{19,44}$ $\therefore \hat{Q}_1 = 57,16^\circ$ $\therefore \theta = 32,84^\circ$	✓ for sub. in the cosine rule <i>verv. in die cosinus-reël</i> ✓ for the answer PQ <i>vir die antwoord PQ</i> ✓ for using the sine rule <i>vir gebruik van die sinus-reël</i> ✓ for/vir $\hat{Q}_1$ ✓ for/vir θ	(5)
			[9]

QUESTION 8 / VRAAG 8

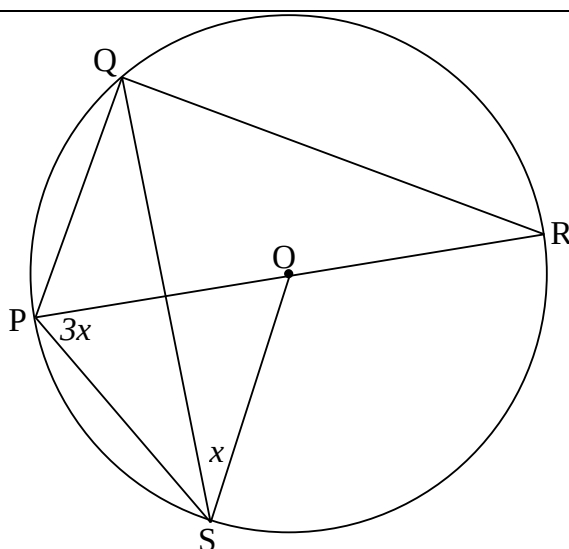


8.1	$a = 2 \quad b = -30^\circ$	✓ for/vir a ✓ for/vir b	(2)
8.2	$C(150^\circ; -1)$	✓ for/vir 150° ✓ for/vir -1	(2)
8.3	$-120^\circ \leq x \leq -90^\circ$	✓ for/vir -120° ✓ for/vir -90° ✓ for correct notation vir korrekte notasie	(3)
8.4	$f(x) = \sin 2(x - 30^\circ)$	✓ for correct answer vir korrekte antwoord	(1)
			[8]

QUESTION 9/VRAAG 9

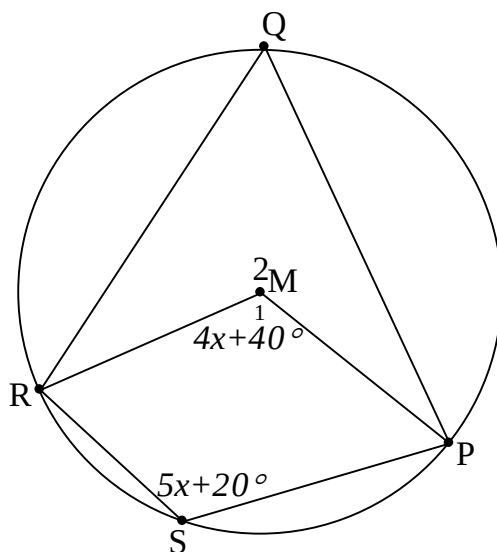


9.1.1	< at centre = 2 x angle at the circumference <i>Middelpunts $\angle = 2 \times \text{Omtreks } \angle$</i>	✓ for the answer <i>vir die antwoord</i>	(1)
9.1.2	<'s on a straight line <i>$\angle$'e op 'n reguitlyn</i>	✓ for the answer <i>vir die antwoord</i>	(1)
9.1.3	Exterior angle of a triangle / <'s of a triangle <i>Buitehoek van 'n driehoek / $\angle$e van 'n driehoek</i>	✓ for the answer <i>vir die antwoord</i>	(1)
9.1.4	$\hat{A} = \hat{F}_1$; corresponding angles = ; EA BOF <i>ooreenkomstige $\angle$e = ; EA BOF</i>	✓ for the answer <i>vir die antwoord</i>	(1)
9.1.5	alternate angles = ; EA BOF <i>verwisselende hoeke = ; EA BOF</i>	✓ for the answer <i>vir die antwoord</i>	(1)



9.2.1	$\widehat{SQR} = 3x$ ($\angle$ s in the same segment) ($\angle$ e in dieselfde segment)	✓ S ✓ R	(2)
9.2.2	$\widehat{Q} = 90^\circ$ ($\angle$ s in a semi-circle) ($\angle$ e in 'n halwe-sirkel) $\therefore \widehat{PQS} = 90^\circ - 3x$	✓ S ✓ R ✓ answer / antwoord	(3)
9.2.3	$\widehat{PSO} = 3x$ ($\angle$ s opposite equal sides) ($\angle$ e teenoor gelyke sye) $\therefore \widehat{PSQ} = 2x$ ($\widehat{QSO} = x$)	✓ S ✓ R ✓ answer / antwoord	(3)
9.2.4	$\widehat{PRQ} = 2x$ ($\angle$ s in the same segment) ($\angle$ e in dieselfde segment)	✓ S ✓ R	(2)
9.2.5	$\widehat{QPR} = 180^\circ - (2x + 90^\circ)$ ($\angle$ s of a triangle) ($\angle$ e van 'n driehoek) $= 90^\circ - 2x$	✓ S and/en R ✓ answer / antwoord	(2)
			[17]

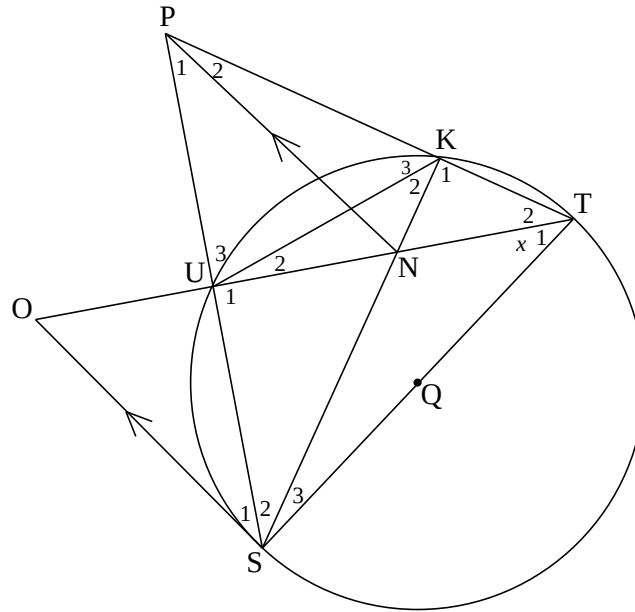
QUESTION 10/VRAAG 10



10.1	$\hat{Q} = 2x + 20^\circ$ ($\angle$ at the centre = $2 \times$ angle at the circumf.) (Middelpunts $\angle$ = $2 \times$ Omtrekshoek) $\therefore 2x + 20^\circ + 5x + 20^\circ = 180^\circ$ (opp. $\angle$s of a c.q.) (teenoorst. $\angle$e van 'n k.v.) $7x + 40^\circ = 180^\circ$ $7x = 140^\circ$ $\therefore x = 20^\circ$ $\therefore \hat{Q} = 60^\circ$ OR / OF $\hat{M}_2 = 360^\circ - (4x + 40^\circ)$ ($\angle$s around a point) ($\angle$e rondom 'n punt) $= 320^\circ - 4x$ $320^\circ - 4x = 2(5x + 20^\circ)$ ($\angle$ at the centre) / (Middelpunts $\angle$) $320^\circ - 4x = 10x + 40^\circ$ $14x = -280^\circ$ $\therefore x = 20^\circ$ $\therefore \hat{Q} = 60^\circ$	$\checkmark$ for/vir S $\checkmark$ for/vir R $\checkmark$ for S and R vir S en R $\checkmark$ for simplifying vereenvoudiging $\checkmark$ the answer die antwoord $\checkmark$ for S and R vir S en R $\checkmark$ for the answer vir die antwoord $\checkmark$ for S and R vir S en R $\checkmark$ for simplifying vereenvoudiging $\checkmark$ for the answer vir die antwoord
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(5)

10.2			
	Construct diameter AE and join C to E. Label $\hat{C}_1$, $\hat{C}_2$ and $\hat{A}_2$ $\hat{C}_1 + \hat{C}_2 = 90^\circ$ ($\angle$ in a semi - circle) $\hat{A}_1 + \hat{A}_2 = 90^\circ$ (tan $\perp$ rad) But $\hat{C}_2 = \hat{A}_2$ ($\angle$s in the same segment) $\therefore \hat{A}_1 = \hat{C}_1$	✓ for construction ✓ for S ✓ for R ✓ for S ✓ for R ✓ for S and R	
	Teken Middellyn AE en verbind C aan E. Merk $\hat{C}_1$, $\hat{C}_2$ en $\hat{A}_2$ $\hat{C}_1 + \hat{C}_2 = 90^\circ$ ($\angle$ in 'n halwe - sirkel) $\hat{A}_1 + \hat{A}_2 = 90^\circ$ (raaklyn $\perp$ radius) Maar $\hat{C}_2 = \hat{A}_2$ ($\angle$e in dieselfde segment) $\therefore \hat{A}_1 = \hat{C}_1$	✓ vir konstruksie ✓ vir S ✓ vir R ✓ vir S ✓ vir R ✓ vir S en R	(6)



10.3.1	$\hat{U}_1 = 90^\circ$ ($\angle$ s in a semi - circle) <i>($\angle$e in 'n halwe - sirkel)</i> $\hat{K}_1 = 90^\circ$ ($\angle$ s in a semi - circle) <i>($\angle$e in 'n halwe - sirkel)</i> $\hat{K}_3 + \hat{K}_2 = 90^\circ$ ($\angle$ s on a straight line) <i>($\angle$e op 'n reguitlyn)</i> $\therefore \hat{U}_1 = \hat{K}_2 + \hat{K}_3 = 90^\circ$ $\therefore$ PUNK is a c.q. (conv. exterior $\angle$ of a c.q.) PUNK is 'n k.v. (omgekeerde Buite $\angle$ van k.v.)	✓ for/vir S and/en R ✓ for/vir S and/en R ✓ for/vir S and/en R ✓ for/vir S ✓ for/vir R	(5)
10.3.2	$\hat{T}_2 = x$ (TO bisects $\hat{S}\hat{T}\hat{P}$) / (TO halveer $\hat{S}\hat{T}\hat{P}$) $\hat{T}_1 = \hat{K}_2 = x$ ($\angle$ s in the same segment) <i>($\angle$e in dieselfde segment)</i> $\hat{K}_2 = \hat{P}_1 = x$ ($\angle$ s in the same segment) <i>($\angle$e in dieselfde segment)</i> $\hat{P}_1 = \hat{S}_1 = x$ (alt. $\angle$ s; $PN \parallel OS$) / (Verw. $\angle$ e ; $PN \parallel OS$) $\therefore \hat{S}_1 = \hat{T}_1 = x$ $\therefore$ SO is a tangent (conv. tan-chord theorem) SO is 'n raaklyn (omgekeerde raaklyn-koord stelling)	✓ for S /vir S/✓ and/en R ✓ for S /vir S /and/en R R ✓ S and/en R ✓ for conclusion vir gevolgtrekking ✓ for R /vir R	(6)
10.3.3	$\hat{T}_1 = \hat{T}_2 = x$ (proven/bewys) $\hat{S}_1 = \hat{T}_2 = x$ (proven/bewys) $\therefore$ POST is a c.q. (conv. $\angle$ s in the same segment) POST is 'n k.v. (omgekeerde $\angle$ e in dieselfde segment)	✓ for/vir S and/en R ✓ for S /vir S and/en R R ✓ for R/vir R	(3)
[25]			

TOTAL/TOTAAL: 150